

**Math 675 Spring 2026:
Project 2 Notes
Due Thursday, May 14th, 2026**

Progress Checklist

- **Introduction:** (Problem, motivation, and context)
Completed
- **Background:** (Definitions and setup)
Completed
- **Model Development:** (Derivation and assumptions)
Almost Done, needs images and more details
- **Analysis:** (Mathematical results and explanations)
In Progress, needs images and more details
- **Discussion:** (Interpretation, limitations, and data considerations)
- **Real World Data Comparison:** (Comparison to actual dataset)
In Progress, explain code, all the variations, what else it's doing and how it's comparing the model to actual dataset, needs more details and images
- **Analysis:** (Mathematical results and explanations)
In Progress, needs images and more details
- **Discussion:** (Interpretation, limitations, and data considerations)
- **Deeper Dive?** (Extending ideas and exploring more complex models)
- **Conclusion:** (Summary and extensions)
- **Reflection:** (Audio Feedback changes)
- **AI Usage Citations:** (Clearly cite AI and explain how it was used)
- **References:** (Literature and sources)
Completed, but should filter out the unnecessary ones
- **Include more images than poster:**
- **Final Formatting Tweaks:**
- **Final Page Count excluding Images:**
- **Summary of work on Presentation:**

Introduction

If someone were to give you along rod that is heated at one end and cooled at the other, and that person told you to provide the temperature at a particular point along the rod at a particular time, you would not be automatically able to calculate that value without some external help or algorithms. We would want to solve a partial differential equation that describes how heat diffuses through the rod, which is called the heat equation, to understand how the temperature changes over time and space.

But because this is a difficult equation to solve analytically and even more complicated to obtain a nice closed form solution, we can use an iterative method to approximate the temperature distribution along the rod at discrete points in space and time. This method allows us to solve this equation, we can use the finite element method (FEM) to discretize the spatial domain and then apply a time-stepping scheme to evolve the solution over time.

FEM is quite similar to a finite difference method, but as we deal with more complex geometries and higher dimensions, the finite element method becomes more advantageous due to its flexibility in handling irregular meshes and complex boundary conditions.

Background

The Governing Heat Equation

For a one-dimensional rod, the heat conduction is governed by:

$$\rho c_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2}$$

- $\frac{\partial T}{\partial t}$ = the rate of change of temperature with respect to time ($^{\circ}C/s$)
- $\frac{\partial^2 T}{\partial x^2}$ = sensitivity of temperature changes at a specific point from its neighbors ($^{\circ}C/m^2$)
- t = Time (s), • x = Position along the rod (m), • $\alpha = \frac{k}{\rho c_p}$ is the thermal diffusivity (m^2/s)
- ρ = Density (kg/m^3) • k = Thermal Conductivity ($W/m \cdot ^{\circ}C$), • c_p = Specific Heat ($J/kg \cdot ^{\circ}C$)

If we substitute in these units into the heat equation, we can verify that the units are consistent:

$$\left(\frac{kg}{m^3} \cdot \frac{J}{kg \cdot ^{\circ}C} \cdot \frac{^{\circ}C}{s} \right) = \left(\frac{W}{m \cdot ^{\circ}C} \cdot \frac{^{\circ}C}{m^2} \right)$$

$$\left(\frac{J}{m^3 \cdot ^{\circ}C \cdot s} \right) = \left(\frac{J}{s \cdot m^3 \cdot ^{\circ}C} \right)$$

This confirms that the units on both sides of the equation are consistent, validating the physical correctness of the heat equation in terms of dimensional analysis. This new unit is called the Volumetric Heat Capacity, which represents the amount of heat required to raise the temperature of a unit volume of the material by one degree Celsius. It is a crucial parameter in understanding how materials respond to thermal energy and is commonly used in heat transfer calculations and thermal analysis.

Project Extension: From 1D to 2D or 3D

$$\text{2D Heat equation: Cartesian} \quad \rho c_p \frac{\partial T}{\partial t} = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

$$\text{2D Heat equation: Polar} \quad \rho c_p \frac{\partial T}{\partial t} = k \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} \right)$$

$$\text{3D Heat equation:} \quad \rho c_p \frac{\partial T}{\partial t} = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right)$$

Model Development

Boundary and Initial Conditions

The model uses the Dirichlet boundary conditions. Meaning the temperature never changes at the boundaries yet allows every other value to vary. The specific conditions are:

$$T(0, t) = 100^\circ\text{F} \quad (\text{Left Boundary})$$

$$T(L, t) = 20^\circ\text{F} \quad (\text{Right Boundary})$$

$$T(x, 0) = T(x) \quad (\text{Initial Temperature Gradient})$$

Finite Element Analysis

The rod is divided into n equally spaced elements of length $h = L/n$. This discretization allows us to approximate the continuous temperature distribution along the rod using a finite number of points. But this results in a system of equations that can be solved using numerical methods.

1: Spatial Discretization (FEM)

For each element along the rod, the local stiffness or local connectivity matrix, $\mathbf{K}^{(e)}$, how each element connects to its neighbors and transfers heat, is defined as:

$$\mathbf{K}^{(e)} = \frac{k}{h} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

2. Transient System (Backward Euler):

To ensure unconditional stability, an implicit scheme is used:

$$(\mathbf{I} - \alpha \Delta t \mathbf{K}') \mathbf{T}^{n+1} = \mathbf{T}^n + \mathbf{b}_{bc}$$

Let's just quickly derive this Backward Euler scheme for our transient heat equation. Starting with the heat equation:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

We can discretize the time derivative using a backward difference approximation and our spatial second derivative using the finite element method from above rather than a finite difference method. The backward difference approximation for the time derivative is:

$$\frac{T^{n+1} - T^n}{\Delta t} \mathbf{I} = \alpha \mathbf{K}^{(e)}$$

Rearranging this gives us to solve for the temperature at the next time step given the current temperature distribution:

$$T^{n+1}\mathbf{I} = T^n\mathbf{I} + \alpha\Delta t\mathbf{K}^{(e)}$$

Now, we can express the spatial second derivative using the finite element method, which leads to a system of equations that can be represented in matrix form as:

$$(\mathbf{I} - \alpha\Delta t\mathbf{K}')\mathbf{T}^{n+1} = \mathbf{T}^n$$

Project Extension: Record calculation time

3. Boundary Treatment (Dirichlet):

Now, accounting for the initial Dirichlet boundary conditions, we can incorporate the boundary conditions into the system of equations by adding a vector $\mathbf{b}_{bc}$, which enforces the fixed temperatures at the boundaries (100°F, 20°F), effectively driving the heat flow toward a linear steady-state where $\mathbf{K}\mathbf{T} = \mathbf{0}$. meaning there is no net heat flow and the temperature distribution is stable.

$$(\mathbf{I} - \alpha\Delta t\mathbf{K}')\mathbf{T}^{n+1} = \mathbf{T}^n + \mathbf{b}_{bc}$$

Analysis and Results of General Model

Real World Data Comparison

Air Temperature Affect

To make this equation more realistic by incorporating ambient air influence, the 1D heat conduction is now governed by:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} - \nu(T - T_\infty)$$

- ν : Convection Coefficient (s^{-1})
- T = Temperature ($^{\circ}\text{C}$),
- T_∞ = Ambient Temperature ($^{\circ}\text{C}$)

If we convert this to a similar backward Euler scheme, we can derive the following scheme:

$$(\mathbf{I} - \alpha\Delta t\mathbf{K}' + \nu\Delta t\mathbf{I})\mathbf{T}^{n+1} = \mathbf{T}^n + \nu\Delta tT_\infty\mathbf{I} + \mathbf{b}_{bc}$$

This now makes the iteration scheme a bit more complicated than before, yet all we've actually done is add more constants to both sides of the equation to involve the ambient temperature.

Project Extension: Record calculation time again with air affect

Using a Real World Dataset

We are going to use a pretty large dataset of temperature measurements along a rod over time, which was collected by students at the Missouri University of Science and Technology. This dataset also provides extra parameters for different types of metals and a setting for including wind moving the air outside of the rod.

Important mention: In the main corner models, we had a similar number of space and time points, but this model has way more time points to compensate for barely any space points

From what we mentioned before, this dataset is unique because the experiment only had 6 points along the rod that measured the temperature, but they had 4000 time steps, which compensates for the few

spatial points and allows us to have a more accurate model with such a high degree of precision.

What is this provided code doing?

First off, Magic. Secondly,

How accurate is our model?

We can compare our model's predictions to real-world data by measuring the temperature at specific points along the rod over time and comparing those measurements to the values predicted by our finite element model. This comparison allows us to validate our model and assess its accuracy in capturing the heat conduction process.

Analysis and Results of Real World Model

Discussion

Conclusions

Reflection (1 Page)

- Include a short reflection addressing:
- What feedback did you receive on your poster?
- What did you improve in the presentation and paper?
- What mathematical aspects did you deepen?
- What did you learn about communicating applied math?

Audio Feedbackn from Poster

1. well executed focus and application
2. stronger applied poster in the class
3. bit of visual clutter (images too big) and more story
4. clear images but print whole poster
5. great font sizes to see from a distance
6. main thing is add more detailed images and less frozen either and hard to identify final result since they look too similar
7. great analysis of heat equation and details behind them and how the method works
8. really like i took olivers suggestion for real world data thats a strong element of the poster and great content i can keep leaning on
9. main take away is clear but modify the first parts of story and last part is great
10. what did we learn feedback, a box for the take away, not obvious for conclusion for reader but great understanding
11. need to show off the references more in poster but pretty good 98

References (1 Page)

General Links for Sources about FEA

- “Why is a Triangular Element Stiffer?” *EnterFEA*.
<https://enterfea.com/why-is-a-triangular-element-stiffer/>
- “The Importance of Mesh Convergence.” *NAFEMS Knowledge Base*.
<https://www.nafems.org/publications/knowledge-base/the-importance-of-mesh-convergence-par>
- “What is Finite Element Analysis and How Does It Really Work?” *Fidelis FEA*.
<https://www.fidelisfea.com/post/what-is-finite-element-analysis-and-how-does-it-really-w>
- “What Is FEA — Finite Element Analysis?” *SimScale Documentation*.
<https://www.simscale.com/docs/simwiki/fea-finite-element-analysis/what-is-fea-finite-elem>
- “What Is Finite Element Analysis?” *MathWorks Discovery*.
<https://www.mathworks.com/discovery/finite-element-analysis.html>
- “Solving Partial Differential Equations with Finite Elements.” *Wolfram Reference*.
<https://reference.wolfram.com/language/FEMDocumentation/tutorial/SolvingPDEwithFEM.html>

Links for Thermal Property Values Estimator Infomation and Code

- “Material thermal properties estimation via a one-dimensional transient convection model.” *Applied Thermal Engineering*, ScienceDirect.
<https://doi.org/10.1016/j.applthermaleng.2020.116230>
- “Data on the Validation to Determine the Material Thermal Properties Estimation.” *Data in Brief*, ScienceDirect.
<https://doi.org/10.1016/j.dib.2021.107577>
- “Python code for 1D k and h value Estimator.” *Zenodo Repository*.
<https://zenodo.org/records/5683312>
- “oneDkhEstimator Required Dataset.” *Mendeley Data*.
<https://data.mendeley.com/datasets/sbcf36976g/1>